

What **psychiatric medication** is a common cause of drug-induced **nephrogenic diabetes insipidus**?

Lithium

Treat with amiloride

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[...]

Polyuric disorders			
- Defined by excessive excretion of dilute urine, often accompanied by increased thirst			
- Urine typically has a specific gravity < 1.006 and an osmolality < 300 mOsm/kg			
	Primary polydipsia	Central diabetes insipidus	Nephrogenic DI
		AKA Arginine vasopressin deficiency	AKA Arginine vasopressin resistance
Pathophysiology	Excessive water intake	↓ ADH production	Renal resistance to ADH
Seen in	Psychiatric disorders (e.g., schizophrenia), hypothalamic lesions affecting thirst center*	Idiopathic, brain injury (trauma, hypoxia, tumor, surgery, infiltrative diseases)	Congenital, drugs (e.g., lithium, demeclocycline), hypercalcemia, hypokalemia
Serum osmolality	↓	↑	↑
Urine osmolality	↓	↓	↓
Serum sodium	Low to normal	Normal to ↑	Normal to ↑
ADH (vasopressin) levels	Low or normal	Low	Normal or high
Water deprivation test†	Significant ↑ in urine osmolality (normal response)	No change or slight ↑ in urine osmolality	No change or slight ↑ in urine osmolality
Response to desmopressin	No response	Significant ↑ in urine osmolality	Minimal change in urine osmolality
Treatment	Behavioral (fluid restriction)	Desmopressin (DDAVP)	Manage the underlying cause: thiazides, amiloride, indomethacin, low-salt diet

*Polydipsia may be transient if damage is below hypothalamus; median eminence or in the posterior pituitary, as ADH in hypothalamus can still be secreted systemically as posterior pituitary releases median eminence.

†After withholding water for 2-3 hours, urine volume and osmolality should be measured hourly, along with plasma sodium concentration and osmolality.

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Illustration by: [Rajesh Mohanram](#)

What **type** of **diabetes** is associated with **lithium use**?

Nephrogenic diabetes insipidus

Lithium antagonizes the action of ADH in the collecting tubules

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One **class** of drugs that may treat **nephrogenic diabetes insipidus** is **thiazide diuretics**, which enhances the uptake of NaCl and H₂O in the **proximal tubule**

Confusing mechanism: thiazide diuretics decrease reabsorption of NaCl in the early DCT → increased urinary excretion of Na⁺ → decreased ECF volume → increased RAAS activation → increased proximal Na⁺ and H₂O reabsorption → decreased urine output

One **class** of drugs that may treat **nephrogenic diabetes insipidus** is [...], which enhances the uptake of NaCl and H₂O in the **proximal tubule**

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Which treatment for **SIADH** may end up causing **nephrogenic diabetes insipidus**?

Demeclocycline

Demeclocycline is an ADH antagonist

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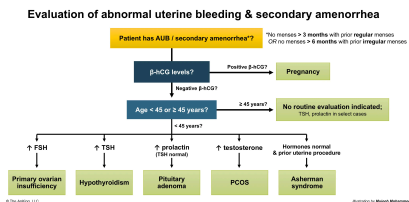
In females, **prolactinoma** presents as **galactorrhea**, **amenorrhea**, and **infertility**

- ↑ Prolactin → ↑ lactogenesis → galactorrhea

- ↑ Prolactin → GnRH inhibition → amenorrhea, infertility

5.1

In females, **prolactinoma** presents as [...], **amenorrhea**, and **infertility**



What is the most common type of **pituitary adenoma**?

Prolactinoma

Adenoma of lactotrophs cause hyperprolactinemia

6.1

What is the most common type of **pituitary adenoma**?

[...]

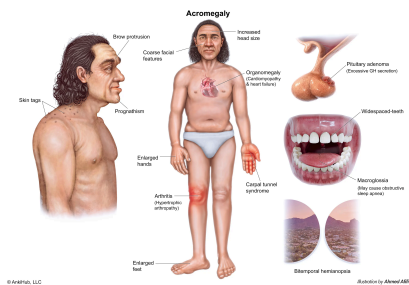
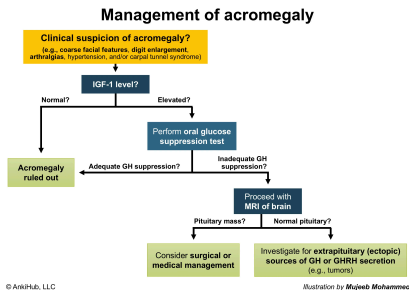
Primary Brain Tumors				
Tumor	Description	Clinical Features	Histology	Other
Pituitary adenoma	Benign slow-growing & non-invasive tumor from adenohypophysis	Varying clinical features depending on size and location	Stains GFAP + Immunohistochemistry (IHC) for Prolactin	Associated with GH type 1
Meningioma	Benign tumor from meninges	May cause symptoms depending on location	Stains GFAP + IHC for Prolactin	Also called meningioma
Ependymoma	Benign tumor from ependymal cells lining ventricles and central canal of spine	May cause symptoms depending on location	Stains GFAP + IHC for Prolactin	Associated with GH type 2
Oligodendroglioma	Benign tumor from oligodendrocytes	May cause symptoms depending on location	Stains GFAP + IHC for Prolactin	Associated with GH type 2
Glioblastoma	Highly malignant tumor from astrocytes	General CNS tumor symptoms	Stains GFAP + IHC for Prolactin	Most common primary brain tumor
Pituitary adenoma	Benign tumor from adenohypophysis	May cause symptoms depending on location	Stains GFAP + IHC for Prolactin	Associated with GH type 1
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Acromegaly is associated with increased **insulin** resistance

Thus contributing to secondary diabetes mellitus

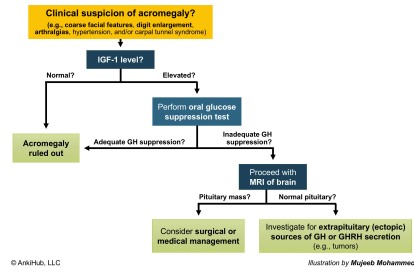
7.1

Acromegaly is associated with increased [...]
resistance

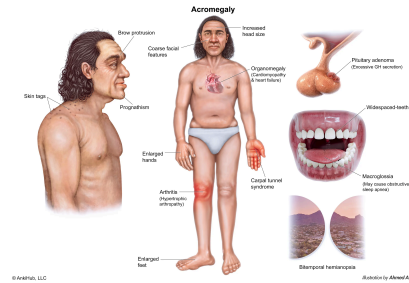


Acromegaly causes impaired **glucose** tolerance (i.e., insulin resistance), which may present as secondary **diabetes mellitus**

Management of acromegaly



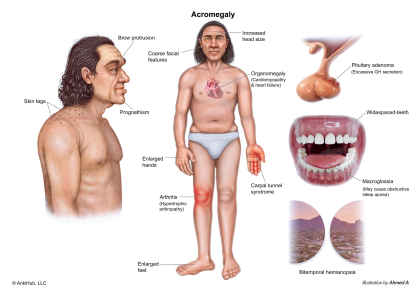
Acromegaly causes impaired [...] tolerance (i.e., insulin resistance), which may present as secondary [...]



Growth hormone-secreting adenoma presents as **acromegaly** in adults

- Acromegaly = Adults

- As opposed to gigantism in children



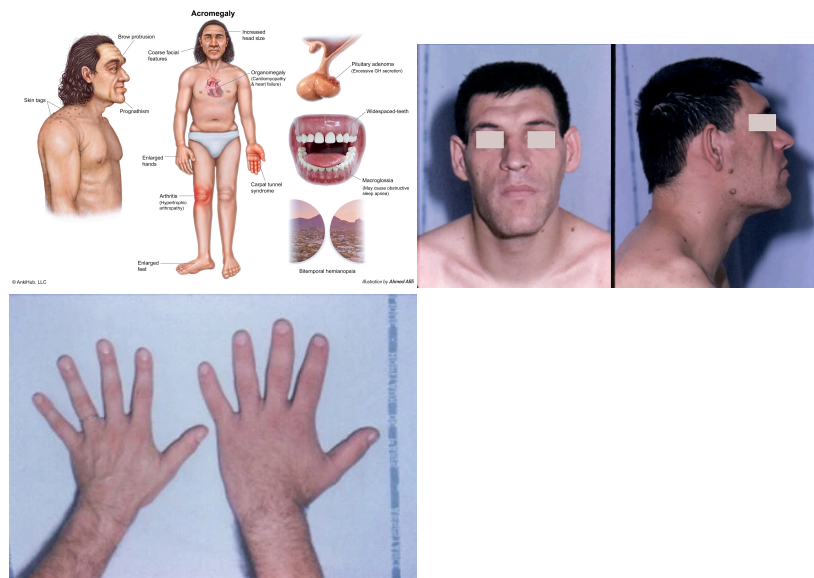
Growth hormone-secreting adenoma presents as [...] in adults



Acromegaly causes enlarged bones in the **hands**, **feet**, and **jaw**

10.1

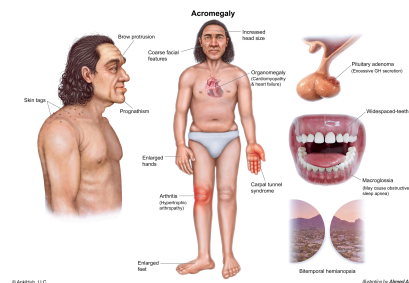
Acromegaly causes enlarged bones in the [...], [...], and [...]



Acromegaly may cause growth of **visceral organs**, leading to dysfunction

11.1

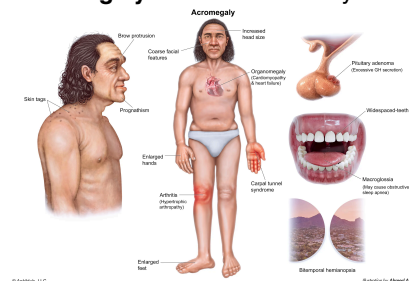
Acromegaly may cause growth of **visceral [...]**, leading to dysfunction



Acromegaly is characterized by a **large tongue** with **deep furrows** and a deep voice

12.1

Acromegaly is characterized by a **large [...]** with **deep furrows** and a deep voice



What is the most common cause of death in patients with **acromegaly**?

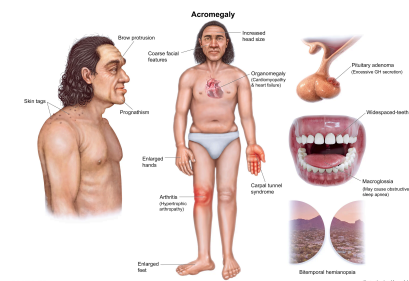
Heart failure

Causes left ventricular hypertrophy

13.1

What is the most common cause of death in patients with **acromegaly**?

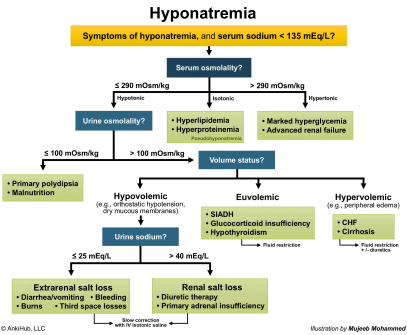
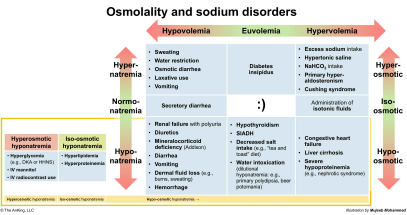
[...]



SIADH is characterized by **euvolemic hyponatremia** with continued urinary Na^+ excretion
Body responds to water retention with **decreased aldosterone**, causing normal volumes (euvolemic) but decreased sodium (hyponatremia)

14.1

SIADH is characterized
by **[volume] [Na⁺ level]**
with continued urinary
 Na^+ excretion



In **diabetes insipidus**, the **water deprivation test** *fails* to **increase** urine osmolality

Osmolality will increase in **central** diabetes insipidus with administration of an **ADH analog**

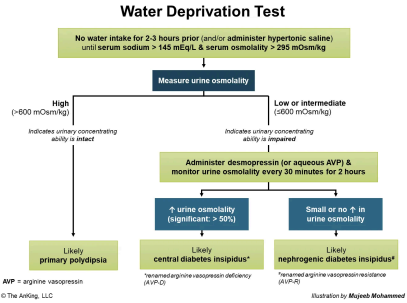
15.1

In **diabetes insipidus**,
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*For withholding water for 8-10 hours, urine volume and osmolality should be measured hourly, along with plasma sodium concentration and osmolality.



Nephrogenic diabetes insipidus is characterized by **polyuria** and **polydipsia** with an inability to **concentrate** urine due to **lack of response** to **ADH**

Two subtypes:

- **Complete nephrogenic diabetes**: complete unresponsiveness of the kidneys to ADH
- **Partial nephrogenic diabetes**: slow but steady rise in urine osmolality with increasing serum osmolality after water deprivation. DDAVP still fails to increase urine osmolality. The urine osmolality remains low (< 500 mOsm/L)

- Whereas **central** diabetes insipidus is due to lack of ADH

16.1

[...] **diabetes insipidus** is characterized by **polyuria** and **polydipsia** with an inability to **concentrate** urine due to **lack of response** to **ADH**

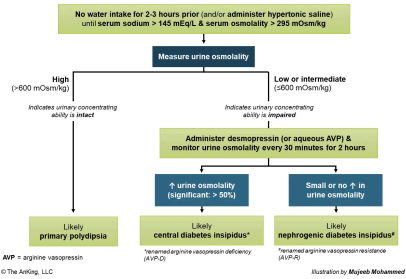
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Water Deprivation Test

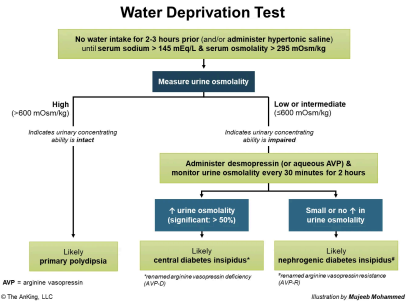


Diagnosis of **diabetes insipidus** is aided by the **water deprivation test**

Alternatively, plasma **copeptin** levels (reflects circulating ADH levels)

17.1

Diagnosis of **diabetes insipidus** is aided by the [...] **test**



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Treatment for **central diabetes insipidus** includes **hydration** and **desmopressin (DDAVP)**, which is an **ADH analog**

18.1

Treatment for **central diabetes insipidus** includes **hydration** and [...], which is an **ADH analog**

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Nephrogenic diabetes insipidus is often caused by **ADH receptor** mutations

Or mutations in genes encoding aquaporins

19.1

Nephrogenic diabetes insipidus is often caused by [...] mutations

20.1

Both **central** and **nephrogenic diabetes insipidus** are characterized by [...] urine osmolality

21.1

Nephrogenic diabetes insipidus is characterized by [...] or [...] levels of ADH

22.1

[...] diabetes insipidus is characterized by **polyuria** and **polydipsia** with an inability to concentrate urine due to **lack of ADH**

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Illustration by: Rajesh Mithunavel

Both **central** and **nephrogenic diabetes insipidus** are characterized by **low** urine osmolality

- Usually < 300 mOsm/Kg
- Due to lack of water reabsorption from absent ADH or ADH resistance in kidneys

Polyuric disorders

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Nephrogenic diabetes insipidus is characterized by **normal** or **elevated** levels of ADH

- Due to fully functional CNS
- Whereas **central** diabetes insipidus is characterized by **low** levels of ADH

Polyuric disorders

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Treatment	Behavioral (fluid restriction)	Desmopressin (DDAVP)	Manage the underlying cause: thiazides, amiloride, indomethacin, low-salt diet

*Polydipsia may be treated if damage to below hypothalamic median eminence or in the posterior pituitary, as ADH in hypothalamus can still be secreted systemically as posterior pituitary is intact.
†After withholding water for 3-5 hours, urine volume and osmolality should be measured hourly, along with plasma sodium concentration and osmolality.
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Illustration by: Rajesh Mithunavel

Central diabetes insipidus is characterized by **polyuria** and **polydipsia** with an inability to concentrate urine due to **lack of ADH**
Whereas **nephrogenic** diabetes insipidus is due to lack of response to ADH

Polyuric disorders

- Defined by excessive excretion of dilute urine, often accompanied by increased thirst
- Urine typically has a specific gravity < 1.006 and an osmolality < 300 mOsm/kg

	Primary polydipsia	Central diabetes insipidus	Nephrogenic DI
		ACA Arginine vasopressin deficiency	ACA Arginine vasopressin resistance
Pathophysiology	Excessive water intake	↓ ADH production	Renal resistance to ADH
Seen in	Psychiatric disorders (e.g., schizophrenia), hypothalamic lesions affecting third ventricle*	Idiopathic, brain injury (trauma, hypoxia, tumor, surgery, infiltrative diseases)	Congenital, drugs (e.g., lithium, demeclocycline), hypercalcemia, hypokalemia
Serum osmolality	↓	↑	↑
Urine osmolality	↓	↓	↓
Serum sodium	Low to normal	Normal to ↑	Normal to ↑
ADH (vasopressin) levels	Low or normal	Low	Normal or high
Water deprivation test†	Significant ↑ in urine osmolality (normal response)	No change or slight ↑ in urine osmolality	No change or slight ↑ in urine osmolality
Response to desmopressin	No response	Significant ↑ in urine osmolality	Minimal change in urine osmolality
Treatment	Behavioral (fluid restriction)	Desmopressin (DDAVP)	Manage the underlying cause: thiazides, amiloride, indomethacin, low-salt diet

*Polydipsia may be treated if damage to below hypothalamic median eminence or in the posterior pituitary, as ADH in hypothalamus can still be secreted systemically as posterior pituitary is intact.
†After withholding water for 3-5 hours, urine volume and osmolality should be measured hourly, along with plasma sodium concentration and osmolality.
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Illustration by: Rajesh Mithunavel

Central diabetes insipidus is characterized by **low** levels of ADH

- Whereas **nephrogenic** diabetes insipidus is characterized by **normal** or **elevated** levels of ADH

- Due to decreased ADH production by the CNS (hypothalamus) → diabetes insipidus (inappropriately dilute urine, polyuria, hypernatremia)

23.1

Central diabetes insipidus is characterized by [...] levels of ADH

Polyuric disorders			
- Defined by excessive excretion of dilute urine, often accompanied by increased thirst			
- Urine typically has a specific gravity < 1.006 and an osmolality < 300 mOsm/kg			
	Primary polydipsia	Central diabetes insipidus	Nephrogenic DI
Pathophysiology	Excessive water intake	↓ ADH production	AKA Arginine vasopressin deficiency
Seen in	Psychiatric disorders (e.g., schizophrenia), hypothalamic lesions affecting thirst center ¹	Idiopathic, brain injury (trauma, hypoxia, tumor, surgery, infiltrative diseases)	AKA Arginine vasopressin resistance
Serum osmolality	↓	↓	↑
Urine osmolality	↓	↓	↓
Serum sodium	Low to normal	Normal to ↑	Normal to ↑
ADH (vasopressin) levels	Low or normal	Low	Normal or high
Water deprivation test ²	Significant ↑ in urine osmolality (normal response)	No change or slight ↑ in urine osmolality	No change or slight ↓ in urine osmolality
Response to desmopressin	No response	Significant ↑ in urine osmolality	Minimal change in urine osmolality
Treatment	Behavioral (fluid restriction)	Desmopressin (DDAVP)	Manage the underlying cause; thiazides, amiloride, indomethacin, low-salt diet

¹Psychosis may be treated if damage is below hypothalamic median eminence or in the posterior pituitary, as ADH in hypothalamus can still be secreted systemically as posterior pituitary is median eminence.
²After withholding water for 8-10 hours, urine volume and osmolality should be measured hourly, along with plasma sodium concentration and osmolality.
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Both **central** and **nephrogenic** diabetes insipidus are characterized by **urine specific gravity < 1.006**

- Unable to concentrate urine

- Normal urine specific gravity = 1.005 - 1.030

24.1

Both **central** and **nephrogenic** diabetes insipidus are characterized by **urine specific gravity < [...]**

Polyuric disorders			
- Defined by excessive excretion of dilute urine, often accompanied by increased thirst			
- Urine typically has a specific gravity < 1.006 and an osmolality < 300 mOsm/kg			
	Primary polydipsia	Central diabetes insipidus	Nephrogenic DI
Pathophysiology	Excessive water intake	↓ ADH production	AKA Arginine vasopressin deficiency
Seen in	Psychiatric disorders (e.g., schizophrenia), hypothalamic lesions affecting thirst center ¹	Idiopathic, brain injury (trauma, hypoxia, tumor, surgery, infiltrative diseases)	AKA Arginine vasopressin resistance
Serum osmolality	↓	↓	↑
Urine osmolality	↓	↓	↓
Serum sodium	Low to normal	Normal to ↑	Normal to ↑
ADH (vasopressin) levels	Low or normal	Low	Normal or high
Water deprivation test ²	Significant ↑ in urine osmolality (normal response)	No change or slight ↑ in urine osmolality	No change or slight ↓ in urine osmolality
Response to desmopressin	No response	Significant ↑ in urine osmolality	Minimal change in urine osmolality
Treatment	Behavioral (fluid restriction)	Desmopressin (DDAVP)	Manage the underlying cause; thiazides, amiloride, indomethacin, low-salt diet

¹Psychosis may be treated if damage is below hypothalamic median eminence or in the posterior pituitary, as ADH in hypothalamus can still be secreted systemically as posterior pituitary is median eminence.
²After withholding water for 8-10 hours, urine volume and osmolality should be measured hourly, along with plasma sodium concentration and osmolality.
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Both **central** and **nephrogenic** diabetes insipidus are characterized by **serum osmolality > 290 mOsm/kg**

Due to decreased ability to retain H₂O

25.1

Both **central** and **nephrogenic** diabetes insipidus are characterized by **serum osmolality [...] 290 mOsm/kg**

Polyuric disorders			
- Defined by excessive excretion of dilute urine, often accompanied by increased thirst			
- Urine typically has a specific gravity < 1.006 and an osmolality < 300 mOsm/kg			
	Primary polydipsia	Central diabetes insipidus	Nephrogenic DI
Pathophysiology	Excessive water intake	↓ ADH production	AKA Arginine vasopressin deficiency
Seen in	Psychiatric disorders (e.g., schizophrenia), hypothalamic lesions affecting thirst center ¹	Idiopathic, brain injury (trauma, hypoxia, tumor, surgery, infiltrative diseases)	AKA Arginine vasopressin resistance
Serum osmolality	↓	↓	↑
Urine osmolality	↓	↓	↓
Serum sodium	Low to normal	Normal to ↑	Normal to ↑
ADH (vasopressin) levels	Low or normal	Low	Normal or high
Water deprivation test ²	Significant ↑ in urine osmolality (normal response)	No change or slight ↑ in urine osmolality	No change or slight ↓ in urine osmolality
Response to desmopressin	No response	Significant ↑ in urine osmolality	Minimal change in urine osmolality
Treatment	Behavioral (fluid restriction)	Desmopressin (DDAVP)	Manage the underlying cause; thiazides, amiloride, indomethacin, low-salt diet

¹Psychosis may be treated if damage is below hypothalamic median eminence or in the posterior pituitary, as ADH in hypothalamus can still be secreted systemically as posterior pituitary is median eminence.
²After withholding water for 8-10 hours, urine volume and osmolality should be measured hourly, along with plasma sodium concentration and osmolality.
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Pituitary adenoma is a **benign** tumor of the **anterior pituitary cells**

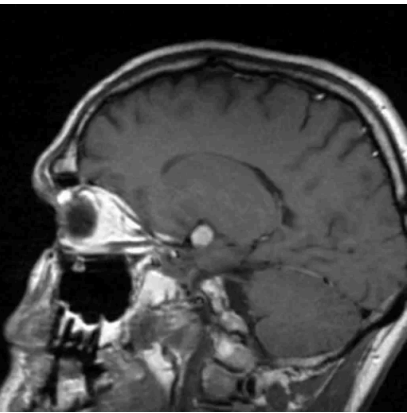
26.1

Pituitary [...] is a **benign** tumor of the **anterior pituitary cells**

Primary Brain Tumors				
Tumor	Description	Clinical Features	Management	Other
Cerebral Tumors	Pilo-cystic astrocytoma	Benign slow-growing & non-invasive tumor (WHO grade I)	Variety of histologic features: Pilocytic astrocytoma	Associated with NF type 1
	Meningioma	Malignant tumor from meninges (WHO grade I)	May cause symptoms: Headache, vomiting, seizures	Also called meningioma, meningioma
	Ependymoma	Malignant tumor from ependymal cells lining ventricles and central canal of brain (WHO grade I)	Paraneoplastic syndrome	Associated with NF type 2
	Oligodendroglioma	Benign tumor from oligodendrocytes (WHO grade I)	Chromosomal changes: loss of 1p and 19q	Dysplastic calcification seen on imaging/histology
	Pilo-cystic astrocytoma	Benign slow-growing & non-invasive tumor (WHO grade I)	Variety of histologic features: Pilocytic astrocytoma	Associated with NF type 1
Anterior Pituitary Tumors	Glucagonoma	High malignant tumor from islet cells (WHO grade I)	Glucagonoma syndrome: hyperglycemia, weight loss, skin lesions	Most common primary endocrine tumor of the pancreas
	Oligodendroglioma	Benign tumor from oligodendrocytes (WHO grade I)	Chromosomal changes: loss of 1p and 19q	Dysplastic calcification seen on imaging/histology
	Meningioma	Malignant tumor from meninges (WHO grade I)	May cause symptoms: Headache, vomiting, seizures	Also called meningioma, meningioma
	Pilo-cystic astrocytoma	Benign slow-growing & non-invasive tumor (WHO grade I)	Variety of histologic features: Pilocytic astrocytoma	Associated with NF type 1
	Schwannoma	Benign tumor from Schwann cells (WHO grade I)	Paraneoplastic syndrome: ataxia, weakness, sensory loss	Associated with NF type 2

Source: WHO Classification of Tumors of the Central Nervous System, 5th Edition (2016).
[1] Pilo-cystic astrocytoma (also called pilocytic astrocytoma) is a slow-growing, non-invasive tumor. [2] Meningioma is a slow-growing, non-invasive tumor. [3] Ependymoma is a slow-growing, non-invasive tumor. [4] Oligodendroglioma is a slow-growing, non-invasive tumor. [5] Pilo-cystic astrocytoma is a slow-growing, non-invasive tumor. [6] Glucagonoma is a malignant tumor. [7] Meningioma is a malignant tumor. [8] Ependymoma is a malignant tumor. [9] Oligodendroglioma is a malignant tumor. [10] Pilo-cystic astrocytoma is a malignant tumor. [11] Schwannoma is a malignant tumor. [12] The Anterior Pituitary Tumors are a group of tumors that arise from the anterior pituitary gland. [13] The Anterior Pituitary Tumors are a group of tumors that arise from the anterior pituitary gland. [14] The Anterior Pituitary Tumors are a group of tumors that arise from the anterior pituitary gland. [15] The Anterior Pituitary Tumors are a group of tumors that arise from the anterior pituitary gland.

One possible cause of hypopituitarism in children is **craniopharyngioma** (due to mass effect)



27.1 One possible cause of hypopituitarism in children is [...] (due to mass effect)

Photo credit: [Hellerhoff](#), [CC BY 3.0](#), via Wikimedia Commons

Primary Brain Tumors					
	Tumor	Description	Clinical Features	Histology	Other
Childhood Tumors	Pilocytic astrocytoma	Benign slow-growing & non-invasive tumor from astrocytes (WHO grade I)	Ventricles Fusiform Pilo-astrocytic	Starry clefts Eosinophilic, eosinophilic fibrils, Rosenthal fibers	Associated with NF type 1
	Meningeothelioma	Malignant tumor from meningeothelial cells of the choroid plexus	May cause hydrocephalus	Stains immunohistochemically + Ki-67 stain with Ki-67 stain	Also called pineal, neuroblastoid tumor
	Ependymoma	Malignant tumor from ependymal cells along ventricular surfaces and central canal of brain	May cause hydrocephalus	Perivascular pseudorosettes Ependymal rosettes	Associated with NF type 2
	Cerebellar astrocytoma	Benign tumor from astrocytes in cerebellum	Stenopteron histopathology	Chondroid nodules found in a tumor of the cerebellum	Dystrophic calcification seen on imaging/histology
	Pinealoma	Tumor of pineal gland	May cause Parinaud syndrome, obstructive hydrocephalus, and/or precocious puberty	Large eosinophilic cells with eosinophilic (pink) granules	
Adult Tumors	Glioblastoma multiforme (GBM)	Highly malignant tumor from astrocytes	General CNS tumor symptoms	Stains GFAP + Pseudopalisading Microvascular proliferation	Most common primary malignant CNS tumor in adults
	Oligodendroglioma	Malignant tumor from oligodendrocytes	May present with seizures	Stains GFAP + Faintly eosinophilic cells with perinuclear halos	Highly vascularized tumor in white matter of cerebral cortex
	Meningioma	Benign tumor from meningeothelial cells of the meninges	Asymptomatic or present with general symptoms	Stains GFAP + Ki-67 stain with Ki-67 stain	Associated with NF type 2
	Pituitary adenoma	Benign tumor of anterior pituitary gland	Hypersecretion of anterior pituitary hormones, causing hypopituitarism, and/or precocious puberty	Stains GFAP + Ki-67 stain with Ki-67 stain	
	Schwannoma	Benign tumor that arises from Schwann cells	Symptoms vary depending on location, sensory hearing loss common	Stains GFAP + Ki-67 stain with Ki-67 stain	Associated with NF type 2

Source: [StatPearls](#), <

One possible cause of hypopituitarism is **pituitary apoplexy**, which is a **sudden hemorrhage** of the pituitary gland, often in the presence of an existing adenoma

Sheehan syndrome is postpartum necrosis of the pituitary gland often following postpartum hemorrhage

28.1 One possible cause of hypopituitarism is [...], which is a **sudden hemorrhage** of the pituitary gland, often in the presence of an existing adenoma

Primary Brain Tumors					
	Tumor	Description	Clinical Features	Histology	Other
Childhood Tumors	Piloicytoma	Benign slow-growing & non-invasive tumor from astrocytes	Younger Ventricles Fusiform cells	Starry clefts Eosinophilic, eosinophilic fibrils, Rosenthal fibers	Associated with NF type 1
	Meningeothelioma	Malignant tumor from meningeothelial cells of the choroid plexus	May cause hydrocephalus	Stains immunohistochemically + Ki-67 stain with Ki-67 stain	Also called pineal, neuroblastoid tumor
	Ependymoma	Malignant tumor from ependymal cells along ventricular surfaces and central canal of brain	May cause hydrocephalus	Perivascular pseudorosettes Ependymal rosettes	Associated with NF type 2
	Cerebellar astrocytoma	Benign tumor from astrocytes in cerebellum	Stenopteron histopathology	Chondroid nodules found in a tumor of the cerebellum	Dystrophic calcification seen on imaging/histology
	Pinealoma	Tumor of pineal gland	May cause Parinaud syndrome, obstructive hydrocephalus, and/or precocious puberty	Large eosinophilic cells with eosinophilic (pink) granules	
Adult Tumors	Glioblastoma multiforme (GBM)	Highly malignant tumor from astrocytes	General CNS tumor symptoms	Stains GFAP + Pseudopalisading Microvascular proliferation	Most common primary malignant CNS tumor in adults
	Oligodendroglioma	Malignant tumor from oligodendrocytes	May present with seizures	Stains GFAP + Faintly eosinophilic cells with perinuclear halos	Highly vascularized tumor in white matter of cerebral cortex
	Meningioma	Benign tumor from meningeothelial cells of the meninges	Asymptomatic or present with general symptoms	Stains GFAP + Ki-67 stain with Ki-67 stain	Associated with NF type 2
	Pituitary adenoma	Benign tumor of anterior pituitary gland	Hypersecretion of anterior pituitary hormones, causing hypopituitarism, and/or precocious puberty	Stains GFAP + Ki-67 stain with Ki-67 stain	
	Schwannoma	Benign tumor that arise from Schwann cells	Symptoms vary depending on location, sensory hearing loss common	Stains GFAP + Ki-67 stain with Ki-67 stain	Associated with NF type 2

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The most common cause of hypopituitarism in adults is **pituitary adenoma** (due to mass effect)

Primary Brain Tumors					
	Tumor	Description	Clinical Features	Histology	Other
Childhood Tumors	Piloicytic astrocytoma	Benign slow-growing & non-invasive tumor from astrocytes (WHO grade I)	Classical features	Starry (GFAP +) Eosinophilic, eosinophilic fibrils, Rosenthal fibers	Associated with NF type 1
	Meningeothelioma	Malignant tumor from meningeothelial cells of the choroid plexus	May cause hydrocephalus	Stains immunohistochemically + Ki-67 stain with Ki-67 stain	Also called pineal, neuroblastoid tumor
	Ependymoma	Malignant tumor from ependymal cells along ventricular surfaces and central canal of brain	May cause hydrocephalus	Perivascular pseudorosettes Ependymal rosettes	Associated with NF type 2
	Cerebellar astrocytoma	Benign tumor from astrocytes in cerebellum	Stenopteron histopathology	Chondroid nodules found in a tumor of the cerebellum	Dystrophic calcification seen on imaging/histology
	Pinealoma	Tumor of pineal gland	May cause Parinaud syndrome, obstructive hydrocephalus, and/or precocious puberty	Large eosinophilic cells with eosinophilic (pink) granules	
Adult Tumors	Glioblastoma multiforme	Highly malignant tumor from astrocytes	General CNS tumor symptoms	Stains (GFAP +) Pseudopalisading Microvascular proliferation	Most common primary malignant CNS tumor in adults
	Oligodendroglioma	Malignant tumor from oligodendrocytes	May present with seizures	Stains GFAP + Faintly eosinophilic cells with perinuclear halos	Highly vascularized tumor in white matter of cerebral cortex
	Meningioma	Benign tumor from meningeothelial cells of the meninges	Asymptomatic or present with general symptoms	Stains GFAP + Ki-67 stain with Ki-67 stain	Associated with NF type 2
	Pituitary adenoma	Benign tumor of anterior pituitary gland	Hypersecretion of anterior pituitary hormones, causing hypopituitarism, and/or precocious puberty	Stains GFAP + Ki-67 stain with Ki-67 stain	
	Schwannoma	Benign tumor that arise from Schwann cells	Symptoms vary depending on location, sensory hearing loss common	Stains GFAP + Ki-67 stain with Ki-67 stain	Associated with NF type 2

Starry clefts: characteristic of Piloicytic astrocytoma

Stenopteron histopathology: characteristic of Cerebellar astrocytoma

Chondroid nodules: characteristic of Cerebellar astrocytoma

Pseudopalisading: characteristic of Glioblastoma multiforme

Microvascular proliferation: characteristic of Glioblastoma multiforme

Perivascular pseudorosettes: characteristic of Ependymoma

Ependymal rosettes: characteristic of Ependymoma

Parinaud syndrome: characteristic of Pinealoma

Precocious puberty: characteristic of Pinealoma

Hypersecretion of anterior pituitary hormones: characteristic of Pituitary adenoma

Hypopituitarism: characteristic of Pituitary adenoma

Sensory hearing loss: characteristic of Schwannoma

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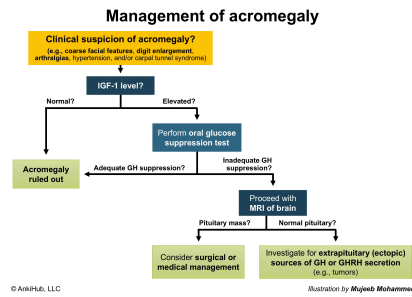
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29.1 The most common cause of hypopituitarism in adults is [...] (due to mass effect)

Growth hormone-secreting adenoma may be diagnosed by elevated serum levels of **growth hormone** and/or **IGF-1**

36.1

Growth hormone-secreting adenoma may be diagnosed by elevated serum levels of [...] and/or [...]



37.1

SIADH is characterized by *increased* [...] of free water

SIADH is characterized by *increased* **retention** of free water

38.1

Hypopituitarism may be caused by [...] **syndrome**, which may occur due to trauma or congenital defect of the sella

Hypopituitarism may be caused by **empty sella syndrome**, which may occur due to trauma or congenital defect of the sella

Image shows empty sella filled with CSF with thin rim of pituitary gland along wall of the cavity

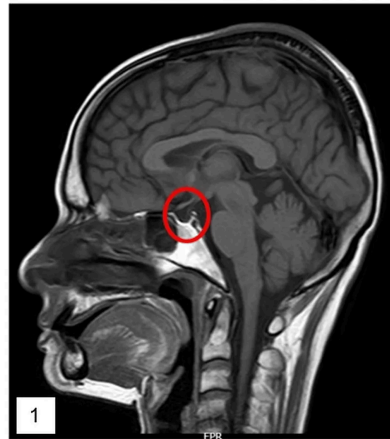


Photo credit: [Ahamed et al., CC BY 4.0](#), via Cureus

39.1

What demographic is more commonly affected by **empty sella syndrome**?
[...]

What demographic is more commonly affected by **empty sella syndrome**?
Obese females

40.1

During **pregnancy**, the size of the **pituitary gland** [#], but blood supply does not increase significantly

During **pregnancy**, the size of the **pituitary gland** **doubles**, but blood supply does not increase significantly
Thus, increased risk of infarction (**Sheehan syndrome**)

41.1

Pregnancy-related infarction of the pituitary gland (**Sheehan syndrome**) often occurs following [...]

Pregnancy-related infarction of the pituitary gland (**Sheehan syndrome**) often occurs following **postpartum hemorrhage** (PPH)
Blood loss increases the risk of infarction

42.1

Pituitary apoplexy presents with sudden onset severe [...], [...] impairment, and features of **hypopituitarism**

Pituitary apoplexy presents with sudden onset severe **headache**, **visual** impairment, and features of **hypopituitarism**
- e.g., bitemporal hemianopsia, diplopia
- This is an **emergency** and needs to be visualized STAT with a **CT**

Prolactinoma may be treated with a(n) **dopamine agonist**, such as **cabergoline** (preferred) or **bromocriptine**

- Also may be treated with surgical removal of the tumor

- Cabergoline preferred over bromocriptine due to efficacy and side-effect profile per AMBOSS (Jun 2023)

Prolactinoma may be treated with a(n) [...] **agonist**, such as **cabergoline** (preferred) or **bromocriptine**

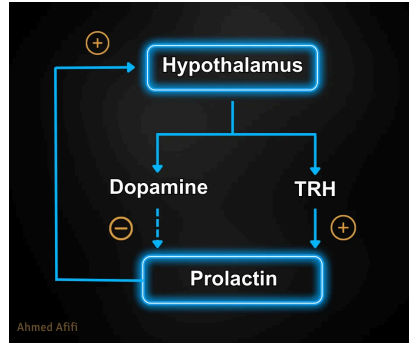


Photo credit: Ahmed Afifi license CC

Growth hormone-secreting adenoma presents as [...] in children

Growth hormone-secreting adenoma presents as **gigantism** in children

- Due to increased linear bone growth

- As opposed to acromegaly in **adults**

Growth hormone-secreting adenoma may be diagnosed by lack of **GH** [...] by oral glucose

Growth hormone-secreting adenoma may be diagnosed by lack of **GH suppression** by oral glucose

The *very low* serum **Na⁺** concentration in **SIADH** can lead to **cerebral** [...] and **seizures**

The *very low* serum **Na⁺** concentration in **SIADH** can lead to **cerebral edema** and **seizures**

- Seizures due to shift of water into neurons

- Hyponatremia due to excess reabsorption of free water

- Can be seen in any other condition that causes hyponatremia

SIADH is characterized by **[serum or urine]** osmolality > **[serum or urine]** osmolality

SIADH is characterized by **urine** osmolality > **serum** osmolality
Inappropriately concentrated urine > 100mOsm/kg due to RAAS, BNP, and reabsorption of water via ADH

[...] is the insufficient production of hormones by the **anterior pituitary gland**

Hypopituitarism is the insufficient production of hormones by the **anterior pituitary gland**

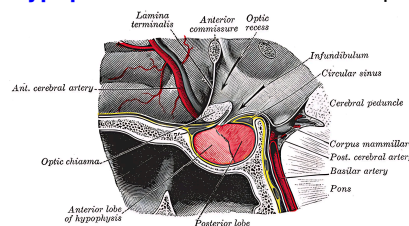


Photo credit: [Henry Vandyke Carter](#), Public domain, via Wikimedia Commons

What is the treatment for **hypopituitarism**?

What is the treatment for **hypopituitarism**?

[...]

Hormone replacement therapy (HRT)

e.g., corticosteroids, thyroxine, sex steroids, human growth hormone

Empty sella syndrome is characterized by **absence** of the **pituitary gland** on imaging

- Often asymptomatic

- Image shows empty sella filled with CSF with thin rim of pituitary gland along wall of the cavity

Empty sella syndrome is characterized by [...] of the **pituitary gland** on imaging



Photo credit: [Ahamed et al.](#), [CC BY 4.0](#), via Cureus, modified by cropping

Symptoms of **Sheehan syndrome** include poor [...], loss of [...], [...]**tension**, and **fatigue**

Symptoms of **Sheehan syndrome** include poor **lactation**, loss of **pubic hair**, **hypotension**, and **fatigue**

- Decreased anterior pituitary function → decreased LH secretion → decreased pubic hair growth

- May also have **absent menstruation** and **cold intolerance**

External causes of hypopituitarism include [...] and **radiation**

External causes of hypopituitarism include **brain injury** and **radiation**

Empty sella syndrome occurs due to atrophy or compression, and ultimately destruction of the **pituitary gland**

- For example, herniation of the arachnoid and CSF into the sella

- Below MRI shows empty sella filled with CSF with thin rim of pituitary gland along wall of the cavity

Empty sella syndrome occurs due to atrophy or compression, and ultimately destruction of the [...] **gland**

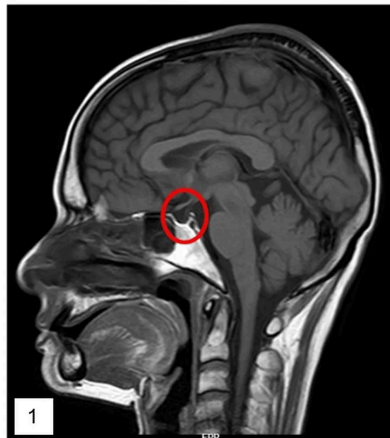


Photo credit: [Ahamed et al.](#), [CC BY 4.0](#), via Cureus

TSH-producing, **LH**-producing, and **FSH**-producing **pituitary adenomas** occur **[frequency]**

TSH-producing, **LH**-producing, and **FSH**-producing **pituitary adenomas** occur **rarely**
Lactotroph adenomas (prolactinomas) are most common

55.1	Symptoms of hypopituitarism arise when > [...] % of the pituitary parenchyma is lost	Symptoms of hypopituitarism arise when > 75% of the pituitary parenchyma is lost
56.1	[...] may be caused by drugs, which may be remembered with the mnemonic "Can't Concentrate Serum Na⁺" [Ca] [C] [S] [N]	SIADH may be caused by drugs, which may be remembered with the mnemonic "Can't Concentrate Serum Na⁺" Carbamazepine Cyclophosphamide SSRIs NSAIDs Also commonly seen with valproic acid
57.1	SIADH may be caused by drugs, which may be remembered with the mnemonic "[...]"	SIADH may be caused by drugs, which may be remembered with the mnemonic "Can't Concentrate Serum Na⁺" Carbamazepine Cyclophosphamide SSRI NSAIDs Also commonly seen with valproic acid
58.1	In <u>males</u>, prolactinoma presents as headache, decreased [...], and infertility	In <u>males</u>, prolactinoma presents as headache, decreased libido, and infertility - Due to inhibition of GnRH release and mass effect; no galactorrhea - HOWEVER, men with prolactinomas are more likely to present at an advanced stage with mass-effect symptoms
59.1	Pituitary adenomas may be [...] (hormone producing) or [...] (silent)	Pituitary adenomas may be functional (hormone producing) or nonfunctional (silent) If functioning, results in hyper___emia of whatever the hormone cell tumor produces (e.g., lactotroph → hyperprolactinemia)

Pituitary adenoma may present with **bitemporal hemianopsia** due to compression of the **optic chiasm**

- Tumor sits in sella turcica, which is adjacent to optic chiasm
- These patients will present with a history of **clumsiness** and **visual changes**

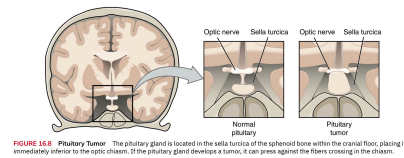


FIGURE 16.8 Pituitary Tumor The pituitary gland is located in the sella turcica of the sphenoid bone within the cranial floor, placing it immediately inferior to the optic chiasm. If the pituitary gland develops a tumor, it can press against the fibers crossing in the chiasm.

Photo credit: [OpenStax](#), [CC BY 4.0](#)

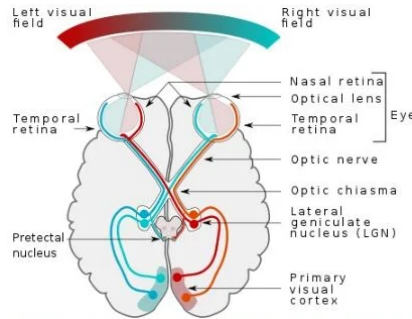


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Primary Brain Tumors

Tumor	Description	Clinical Features	Diagnosis	Other
Pituitary adenoma	Benign, slow-growing, non-invasive tumor from pituitary gland	Visual field defects, hormonal dysfunction	Imaging (MRI, CT), endocrine tests	Associated with NF type 1
Meningioma	Malignant tumor from meninges (outermost layer of brain)	Headaches, seizures, focal neurological deficits	Imaging (MRI, CT)	Also called meningioma; associated with NF type 2
Ependymoma	Malignant tumor from ependymal cells (innermost layer of brain)	Headaches, seizures, focal neurological deficits	Imaging (MRI, CT)	Associated with NF type 2
Oligodendroglioma	Benign tumor from oligodendrocytes (glial cells)	Headaches, seizures, focal neurological deficits	Imaging (MRI, CT)	Associated with NF type 2
Pilocytic astrocytoma	Benign tumor from astrocytes (glial cells)	Headaches, seizures, focal neurological deficits	Imaging (MRI, CT)	Associated with NF type 2
Glioblastoma multiforme (GBM)	Highly malignant tumor from astrocytes	Headaches, seizures, focal neurological deficits, rapid growth	Imaging (MRI, CT)	Most common primary brain tumor; associated with NF type 2
Oligodendroglioma	Malignant tumor from oligodendrocytes	Headaches, seizures, focal neurological deficits	Imaging (MRI, CT)	Associated with NF type 2
Neurofibroma	Benign tumor from Schwann cells (nerve sheath)	Headaches, seizures, focal neurological deficits	Imaging (MRI, CT)	Associated with NF type 2
Pituitary adenoma	Benign tumor from pituitary gland	Headaches, seizures, focal neurological deficits	Imaging (MRI, CT)	Associated with NF type 2
Schwannoma	Benign tumor from Schwann cells (nerve sheath)	Headaches, seizures, focal neurological deficits	Imaging (MRI, CT)	Associated with NF type 2

Source: Adapted from: [National Cancer Institute](#). [Brain Tumors](#). NCI Thesaurus. Code C8491. © The Author, LLC. All rights reserved.

Hypopituitarism may be caused by **Sheehan syndrome**, which is a(n) **pregnancy-related infarction** of the pituitary gland

- Pregnancy-induced growth of pituitary (estrogen stimulates hyperplasia of lactotrophs), as well as the minimal vascularity of the pituitary gland, make the pituitary susceptible to hypoperfusion
- Therefore, heavy postpartum bleeding will manifest with hypoperfusion of the pituitary

One common cause of **nephrogenic diabetes insipidus** is **lithium**, which prevents proper aquaporin formation in principal cells

- In the **collecting duct**

- **Lithium-induced nephrogenic DI** is treated with salt restriction and discontinuation of lithium if lithium can be discontinued

Polyuric disorders			
- Defined by excessive excretion of dilute urine, often accompanied by increased thirst			
- Urine typically has a specific gravity < 1.006 and an osmolality < 300 mOsm/kg			
	Primary polydipsia	Central diabetes insipidus	Nephrogenic DI
Pathophysiology	Excessive water intake	ADH deficiency	Renal resistance to ADH
Seen in	Psychiatric disorders (e.g., schizophrenia), hypothalamic lesions affecting thirst center ⁽¹⁾	Idiopathic, brain injury (trauma, hypoxia, tumor, surgery), infiltrative diseases (amyloidosis)	Congenital, drugs (e.g., lithium, demeclocycline), hypercalcemia, hypokalemia
Renal excretion	↓	↓	↓
Urine osmolality	Low	Low	Low
Serum sodium	Low to normal	Normal to ↑	Normal to ↑
ADH (vasopressin) levels	Low or normal	Low	Normal or high
Water deprivation test ⁽²⁾	Significant ↑ in urine osmolality (normal response)	No change or slight ↑ in urine osmolality	No change or slight ↓ in urine osmolality
Response to desmopressin	No response	Significant ↑ in urine osmolality	Minimal change in urine osmolality
Treatment	Behavioral (fluid restriction)	Desmopressin (DDAVP)	Manage the underlying cause; thiazides, acetazolamide, indomethacin, low-salt diet

⁽¹⁾Thirst may be limited if damage to higher hypothalamic median eminence or to posterior pituitary, i.e., ADH is hyposecreted or not secreted adequately.
⁽²⁾Water deprivation test for 2-3 hours; urine volume and osmolality should be measured hourly, along with plasma sodium concentration and osmolality.

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60.1 Pituitary adenoma may present with [...] due to compression of the **optic chiasm**

61.1 Hypopituitarism may be caused by [...] syndrome, which is a(n) **pregnancy-related infarction** of the pituitary gland

62.1 One common cause of **nephrogenic diabetes insipidus** is [...], which prevents proper aquaporin formation in principal cells

Etiologies of [...] diabetes insipidus include pituitary tumor, autoimmune, trauma, surgery, and ischemic encephalopathy

Etiologies of **central diabetes insipidus** include pituitary tumor, autoimmune, trauma, surgery, and ischemic encephalopathy

Polyuric disorders			
- Defined by excessive excretion of dilute urine, often accompanied by increased thirst			
- Urine typically has a specific gravity < 1.006 and an osmolality < 300 mOsm/kg			
	Primary polydipsia	Central diabetes insipidus	Nephrogenic DI
	ACA Arginine vasopressin deficiency	ACA Arginine vasopressin deficiency	ACA Arginine vasopressin resistance
Pathophysiology	Excessive water intake	↓ ADH production	Renal resistance to ADH
Seen in	Psychiatric disorders [e.g., schizophrenia, hypothalamic lesions affecting thirst center]	Idiopathic, brain injury (trauma, hypoxia, tumor, surgery, infiltrative diseases)	Congenital drugs (e.g., lithium, demeclocycline), hypercalcemia, hypokalemia
Serum osmolality	↓	↑	↑
Urine osmolality	↓	↓	↓
Serum sodium	Low to normal	Normal to ↑	Normal to ↑
ADH (vasopressin) levels	Low or normal	Low	Normal or high
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Response to desmopressin	No response	Significant ↑ in urine osmolality	Minimal change in urine osmolality
Treatment	Behavioral (fluid restriction)	Desmopressin (DDAVP)	Manage the underlying cause; thiazides, acetohex, indomethacin, low-salt diet

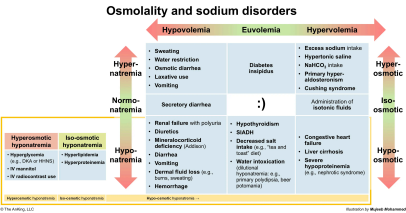
*Polydipsia may be treated if damage to hypothalamic median eminence or in the posterior pituitary, as ADH in hypothalamus can still be secreted systemically as pituitary capillaries in median eminence

*After withholding water for 2-3 hours, urine volume and osmolality should be measured hourly, along with plasma sodium concentration and osmolality

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Diabetes insipidus will cause a(n) [...]osmotic change with [contraction/expansion]

Diabetes insipidus will cause a(n) **hyper**osmotic change with **euvolemia**
Free water loss → increased serum osmolality → water moves from ICF to ECF → contraction of both ECF & ICF; however, **thirst maintains volume status (euvolemia)**



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Serum osmolality	↓	↑	↑
Urine osmolality	↓	↓	↓
Serum sodium	Low to normal	Normal to ↑	Normal to ↑
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Treatment	Behavioral (fluid restriction)	Desmopressin (DDAVP)	Manage the underlying cause; thiazides, acetohex, indomethacin, low-salt diet

*Polydipsia may be treated if damage to hypothalamic median eminence or in the posterior pituitary, as ADH in hypothalamus can still be secreted systemically as pituitary capillaries in median eminence

*After withholding water for 2-3 hours, urine volume and osmolality should be measured hourly, along with plasma sodium concentration and osmolality

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